



Research Paper

Critical heat flux in helical coils at low pressure



B.K. Hardik, S.V. Prabhu*

Department of Mechanical Engineering, Indian Institute of Technology, Bombay, India

HIGHLIGHTS

- Review is presented on critical heat flux in helical coils.
- Axial wall temperature distribution is analysed on inner and outer side of coils.
- Effect of heat flux, mass flux, curvature and quality on CHF is studied.
- CHF is studied in complete saturated region with quality ranging from 0 to 1.
- Simple correlation is developed to predict the critical heat flux at low pressure.

ARTICLE INFO

Article history:

Received 17 August 2016

Revised 14 October 2016

Accepted 19 October 2016

Available online 28 October 2016

Keywords:

Helical coil

Local wall temperature distribution

Critical heat flux

Low pressure

Infrared thermal camera

ABSTRACT

The work presents the study on flow boiling crisis in helical coils at low pressure with water as working medium. Published correlations of critical heat flux in helical coils are collated. The characteristics of boiling crisis in helical coils are studied. The experiments are performed with six helical coils made of SS 304 tubes having inner diameters varying from 6 to 10 mm. The coil diameter to the tube diameter ratio ranges from 14 to 58. The critical heat flux data are collected in complete saturated region for the quality range of 0–1. The wall temperature distribution is measured in the axial direction using infrared thermal imaging technique. The results concluded that, the wall temperature in inner half surface is higher than outer half surface of helical coils. Increase in mass flux increases the critical heat flux. Critical heat flux decreases with increase in quality. For the same quality, the critical heat flux decreases with increase in tube diameter and coil diameter. In a high quality region, the critical heat flux in helical coils is higher than vertical straight tubes. A correlation is suggested for critical heat flux in helical coils.

© 2016 Elsevier Ltd. All rights reserved.

1. Introduction

The critical heat flux phenomenon is studied well for flow inside a straight tube. However, CHF mechanism inside a helical coil may differ due to phase separation with low density vapour being on inner side and high density liquid on outer side of the curved tube or the appearance of thin liquid film on the circumference caused by the gravity and centrifugal forces. Thus, a large temperature distribution may occur on the circumference of tube wall and may affect the heat transfer performance and can cause dryout of liquid film on inner side of curved pipe. Therefore, a detailed study of the wall temperature distribution and CHF characteristics in the helical coil is essential. It is well-known that, during saturated two phase flow boiling, two boiling mechanisms such as nucleate boiling and

convective boiling determine the heat transfer within the tube. In the practical applications, forced convective boiling is the heat-transfer mode, where the highest heat transfer coefficient is achieved at very small temperature differences between tube wall and bulk fluid temperature. With excessive heat supplied, the convective boiling reaches to dryout type critical heat flux. Similarly, during nucleate boiling, the high heat flux creates burnout of tube because of departure from nucleate boiling. Hence, it is of great practical significance to have well-established design recommendations to estimate the critical flux.

Naphon and Wongwises [1] and Vashisth et al. [2] conducted a review on single and two phase flow in curved tubes. They stated that, in spite of wide range of application of curved geometry in industrial devices, there is lack of fundamental knowledge and literature studied on curved tubes compared to that of straight tubes. Recently, Fsadni and Whitty [3,4] presented review on two-phase heat transfer and pressure drop in helical coils. Review includes the literature available on water, R134a and nanofluids. The review work presented by [1–4] on two phase flow is limited to pressure

* Corresponding author at: Department of Mechanical Engineering, Indian Institute of Technology, Bombay, Powai, Mumbai, India.

E-mail addresses: h.kothadia@gmail.com (B.K. Hardik), svprabhu@iitb.ac.in (S.V. Prabhu).

Nomenclature

Symbol

C_p	specific heat at constant pressure [J/kg K]
D	coil diameter [m]
d	tube diameter [m]
G	mass flux [kg/m ² s]
h	enthalpy [J/kg]
I	current [A]
k	thermal conductivity [W/m K]
L	length [m]
$\dot{m}$	mass flow rate [kg/s]
P	pressure [N/m ²]
p	pitch [m]
Q	heat supply [W]
q''	heat flux [W/m ²]
T	temperature [°C]
t	wall thickness [m]
V	voltage [V]
v	velocity [m/s]
x	quality of steam

Greek

μ	dynamic viscosity [N s/m ²]
ρ	density [kg/m ³]

σ surface tension [N/m]

Subscript Definition

1	first dryout
b	bulk
c	coil
fg	liquid to vapor
g	vapor
h	heated
i	inlet
l	liquid
o	outer
sat	saturated
w	wall

Abbreviation

CHF	critical heat flux [W/m ²]
DNB	departure from nucleate boiling

Dimensionless number

Bo	Boiling number $Bo = q''/Gh_{fg}$
Re	Reynolds number $Re = Gd/\mu$

drop and heat transfer. Hence, there is still lack of literature on the review of work done on boiling crises and the present status of knowledge for critical heat flux in helical coils.

Berthoud and Jayanti [5] characterised the dryout in helical coils. They analysed the mechanism based dryout for high pressure system. Unlike the straight tubes, the dryout in helical coils is divided as first dryout, partial dryout and total dryout. The correlations are developed for total dryout. Based on the occurrence of dryout, the first dryout is classified as stratification - redeposition - entrainment dominated dryout. The dryout map as shown in Fig. 1(A) is developed to divide and measure the first dryout in different dominated zones. The influence of system parameters on dryout is studied. Recently, Hwang et al. [6] conducted experiments with three helical coils. They concluded that the secondary flow is not affecting boiling heat transfer in helical coil. Different from that the dryout is significantly affected by secondary flow

and hence coil curvature. Berthoud and Jayanti [5] dryout map is modified for redeposition zone and gravity zone as shown in Fig. 1(B). In the gravity affected zone, the CHF quality in a helical coil is lower than the straight vertical tube, while in redeposition zone, CHF quality is higher than the straight vertical tube.

Jensen and Bergles [7] conducted experiments in straight horizontal tubes and helical coils in subcooled and saturated boiling region. They concluded that subcooled CHF in helical coils is lower than that in straight horizontal tubes. The CHF in saturated region is higher in helical coils. The variation of CHF in a helical coil with the increase in quality measured by Jensen and Bergles [7] is shown in Fig. 2(A). Jensen and Bergles [7] correlates the subcooled CHF in helical coils with CHF in straight horizontal tubes. Because of less number of data for quality CHF in a straight tube, they could not correlate helical CHF with straight horizontal CHF in the saturated region. Jayanti and Berthoud [8] conducted the CHF

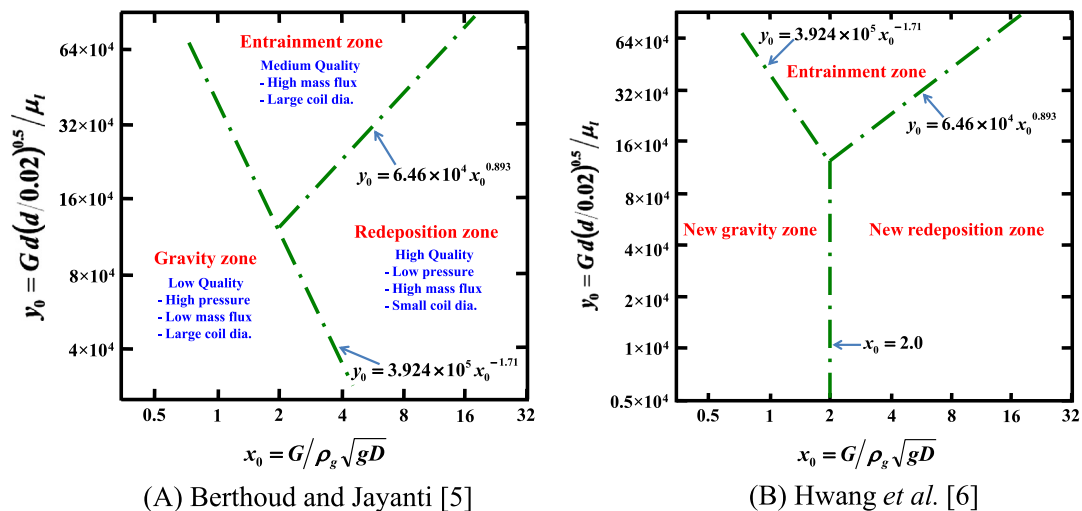


Fig. 1. First dryout prediction map.

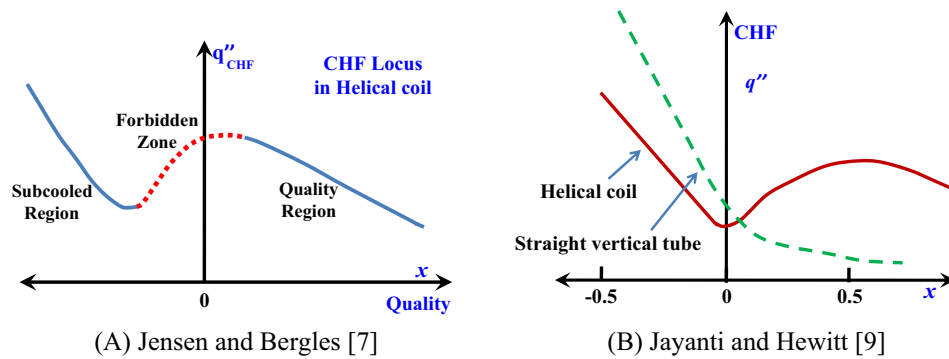


Fig. 2. Variation of CHF with quality in helical coils.

experiments in high quality region. Jayanti and Hewitt [9] studied the subcooled CHF in helical coils. In the subcooled region ($x < 0$), the CHF in coils is less than that in straight tubes, whereas it is higher in the quality region ($x > 0$). Jayanti and Hewitt [9] schematically represent the CHF in helical coils and straight tubes in subcooled and saturated region. The CHF decreases with quality up to a quality of about zero, and then increases to a maximum in the intermediate quality region. CHF decreases with quality in the high quality region. Fig. 2(B) shows that the subcooled CHF in helical coils is lower compared to straight vertical tubes. In a quality region, the CHF in helical coils is much higher than in straight vertical tubes. Jensen and Bergles [7] concluded that helical coil CHF decreases with the increase in quality in the saturated region. However, Jayanti and Hewitt [9] suggest an increase in CHF with quality in the low quality saturated region and then decreasing trend with the increase in the quality in the high quality region.

Summary of the effect of different parameters on CHF studied by different authors is reported in Table 1. Table 1 shows the effect of increase of different parameters on CHF. It is clear from Table 1 that the increase in the quality decreases the CHF. Effect of mass flux and curvature ratio is still not conclusive. Styrikovich et al. [12] and Jensen and Bergles [7] reported increase in mass flux decreases CHF. However, other works reported increase of CHF. Similarly, recent work of Chung et al. [22] reported no effect of mass flux, system pressure and curvature ratio on CHF. Very few works are reported on the effect of curvature ratio on CHF. Santini et al. [23] describe four different methods to detect CHF. Table 2 shows the method used by different authors to detect CHF in helical coils. Most of the available literature on helical coils measures the dryout as a first sudden jump in the wall temperature. While the other literature measures the CHF as sharp increase in the wall

Table 2

Method used by different authors to identify critical heat flux.

Author (Year)	Method used	
	3	4
Kozeki [10]	✓	
Jensen [7]		✓
Unal [11]		✓
Styrikovich [12]	✓	
Jayanti [8]		✓
Ma [13]	✓	
Kaji [14]	✓	
Chung [15]	✓	
Matushi [16]		✓
Ami [17]	✓	✓
Chen [18–20]		✓
Tan [21]		✓
Chung [22]	✓	
Hwang [6]		✓
Santini [23]		✓

temperature with a small increase in the heat flux. This method shows the discontinuity in the curve of wall temperature with heat flux. There is no work reported in the literature to measure the CHF with other two methods.

Definitions of critical heat flux [23]:

1. Heat flux at which the heat transfer coefficient reaches a maximum.
2. Heat flux at which a temperature rises sufficient to burst the pressurized heated tube or to heat part of the tube to red.
3. Heat flux at which a discontinuity occurs in the curve of the temperature of the heated wall versus heat flux.

Table 1

Effect of increase of different parameters on critical heat flux.

Author	Fluid	Mass flux	Quality	System pressure	Coil diameter	Curvature ratio (d/D)
Kozeki [10]	Water	Increase	Decrease	Increase		
Jensen [7]	R113	Increase Decrease				Increase
Unal [11]	Water					
Styrikovich [12]	Water	Decrease		Decrease		
Jayanti [8]	R12					
Ma [13]	Water	Increase		No effect		Increase
Kaji [14]	R113	Increase			Decrease	
Chung [15]	R113	Increase		Decrease	Decrease	
Matushi [16]	W+N ₂					
Ami [17]	N ₂					
Chen [18–20]	R134a	Increase	Decrease	Decrease	Decrease	Increase
Tan [21]	R134a					
Chung [22]	Water	No effect		No effect		No effect
Hwang [6]	Water	Increase		Decrease	Decrease	
Santini [23]	Water	Increase		Decrease		

4. Heat flux at which temperature oscillations first appear in the time-temperature recording.

Detailed literature survey is carried out on critical heat flux. Details of each work along with fluid used in each experiment and corresponding operating and geometric parameters of each experimental work are presented in Table 3. Following are the conclusions that may be drawn from literature survey.

- The effect of quality on CHF is not conclusive.
- CHF data in the low quality region is not available.
- Literature on the critical heat flux in helical coils at low system pressure is not available.
- Simple correlation to estimate the CHF of water in helical coils is not available.

The objective of the present work is to study critical heat flux at low pressure with water as the working fluid. The experimental data for CHF are collected in complete saturated region for the quality range of 0–1. The effect of parameters like tube diameter, coil diameter, heat flux, mass flux and quality on critical heat flux is analysed. The wall temperature distribution on the inner side and outer side of helical coil is analysed. The experiments are performed with six helically coiled test sections made of SS 304 tubes having inner diameters varying from 6 to 10 mm. The coil diameter to the tube diameter ratio ranges from 14 to 58 and coil pitch is 50 mm.

2. Experimental set up

The experimental facility, procedure to perform experiment, equations required to calculate experimental CHF and uncertainty in experimental CHF are described in this section.

2.1. Description of experimental facility

Fig. 3 shows an illustration of the experimental facility. It is an open loop flow system with water as working fluid. The test facility consists of an insulated water tank, magnetically coupled sealless gear pump, ball valves, bypass valves, vertical oriented helical coil and straight horizontal tube and insulated water mixing thermos flask. The gear pumps with a mass flow rate capacity of 360 g/s is driven by a D.C. Motor. The speed of the motor is varied between 0 and 3500 rpm by a motor controller.

The test facility is instrumented with temperature, pressure and flow rate measuring devices. Bulk fluid temperature at the inlet and outlet of the test sections are measured with K-type thermocouples. Three thermocouples are dipped into the insulated water tank to measure the inlet fluid temperature and the outlet bulk fluid temperature is measured with five thermocouples immerse in an insulated thermo flask. All the thermocouples are kept in direct contact with fluid to measure the actual fluid temperature. The wall temperature of test section is measured using Thermoteknix - VisIR 640 Infra-Red thermal camera. Thermal camera measures the intensity of radiation emitted by the test section wall. The intensity depends on the temperature and the emissivity of the surface. The test section is painted with a thin coat of high temperature black board paint to have an emissivity of 0.85. DSLR camera is used to record the process of burnout. The pressure transmitters are used to measure the system pressure at the inlet of the test section. The range of pressure transmitter is 0–5 bar. Electromagnetic mass flow meter with two different ranges adjustment 0–50 g/s and 50–500 g/s is used to measure low flow rate and high flow rate of water accurately. All the test sections are heated electrically by passing DC current through the tube wall maintain-

ing the uniform heat flux condition. Low voltage high current DC power supply is used to supply power through the test section. The rate of power supplied is determined from the measured current and voltage across the coil. The DC current is measured using the power supply digital reading, while the voltage is measured using digital multimeter. These are recorded manually. All the instruments are connected to Data Acquisition system (DAS) which is in turn connected to a computer. Eight thermocouple ports of DAS are connected to thermocouple connections fitted at the open end of the thermocouples. The pressure transmitter and flow meter are connected to DAS through 100 Ω registers to convert current signal (4–20 mA) into readable voltage signal (0–2 V). All the readings are taken after the reaching of steady state.

The geometric details of the six helically coiled tubes tested and operating parameters covered in the present study are listed in Table 4. Three different tube diameters 6 mm, 8 mm and 10 mm with two different curvature ratios are used for analysis of tube diameter and curvature effect. Data for different critical quality is obtained by changing the length of test section for the same tube and coil diameter as given in Table 4. The variables used in the present study are mass flux, inlet subcooling, heated length, tube diameter and coil diameter.

2.2. Data reduction

The heat supplied to the helical coil test sections is calculated from the supplied electric current multiplied by the voltage across the test section as shown in Eq. (1).

$$Q = VI \quad (1)$$

Heat loss from test section at given heat load is subtracted from the heat supply. Theoretical calculation is carried out to calculate convective and radiation heat loss from the helical coil wall surface to atmosphere. The heat flux is calculated from Eqs. (2) and (3).

$$Q_{conv} = Q - Q_{loss} \quad (2)$$

$$q'' = \frac{Q_{conv}}{\pi d L_h} = \frac{Q - Q_{loss}}{\pi d L_h} \quad (3)$$

The liquid properties in a subcooled region are calculated at local bulk fluid temperature. In a saturated region, the properties are calculated at the local saturation temperature. The vapour properties in all calculations are calculated at local saturation temperature. Local bulk fluid temperature distribution is given in [24]. The thermo-physical properties used in the data reduction are calculated from the tabulated values of National Institute of Standards and Technology (NIST). The axial temperature distribution of the test section wall is obtained from thermal images for each configuration. The quality in subcooled and saturated region is calculated from Eqs. (4) and (5). An Eq. (4) gives the quality at specific length of test section.

$$x_{local} = \frac{\frac{Q_{conv}}{m} \frac{L}{L_h} - h_l}{h_{fg}} \quad (4)$$

$$h_l = C_p (T_{sat} - T_i) \quad (5)$$

2.3. Uncertainties in the measured and computed parameters

The uncertainty of different parameters used in the experimental analysis is shown in Table 5. The uncertainties in the tube diameter and coil diameter of the test section are obtained by measuring the values at different locations. The uncertainties of mass flow rate, temperature, and pressure are calculated by comparing the reading of data acquisition system with the value obtained from standard instruments. The uncertainties of heat flux resulted from the accuracy of the ampere meter and volt meter. Total length of

Table 3

Available correlations for critical heat flux.

Authors	Condition	d (mm), D (mm), L (mm), p (mm), t (mm)	$\frac{D}{d}$	q'' (kW/m ²); P (bar) G (kg/m ² sec); T (°C)	Correlation
Kozeki [10]	Water Heated Water; Vertical	$d = 16.1$; $t = 3.1$; $D = 628, 682$	39– 42.4	$P = 5\text{--}20$; $G = 161\text{--}486$; $q'' = 151\text{--}349$	Studied the effect of mass flux on CHF
Jensen [7]	Uniform Heat Flux; R 113; Vertical	$d = 7.44, 7.62$; $t = 0.15, 0.25$; $D = 117.5\text{--}409.6$; $L = 635\text{--}1295$	55, 28.3, 15.4	$P = 9.4$; $G = 570\text{--}5470$; $q'' = 54\text{--}800$	Subcooled $(a/g) = v^2/gD$ $Re_g = Gd/\mu_g$ $\frac{q''_{CHF-coil}}{q''_{CHF-straight}} = K = \frac{0.769(a/g)^{-0.26}}{1}$ for $\frac{(a/g)}{(d/D)} > 10$ for $\frac{(a/g)}{(d/D)} \leq 10$
Unal [11]	Sodium Heated Water; Vertical	$d = 18$; $t = 4$; $L = 40.13, 35.50, 26.67$; $D = 700, 1500$	38.9, 83.3	$P = 147\text{--}202$; $G = 112\text{--}1829$; $x = 0.08\text{--}1$; $q'' = 41\text{--}731$	$Bo = 4.09 \times 10^{-6} \times Re_g^{0.5} (d/D)^{0.17} x^{-0.460}$ for $G \leq 950$ $Bo = 0.97 \frac{a_1 a_2 a_3 a_4 a_5}{a_6 a_7}$ $a_1 = 1 + 3.8 \Delta H$; $a_2 = 0.114 - 0.41 \ln(1 - Pr)$ $a_3 = 1 + 4.59 (\frac{L}{d})^{-1.2}$ $a_4 = 1 + 0.44 [\exp(-0.056 \frac{D}{d}) - \exp(-3 \frac{D}{d})]$ (for first detected dryout) $a_5 = 1 + 0.56 [\exp(-0.011 \frac{D}{d}) - \exp(-3 \frac{D}{d})]$ (for last detected dryout) $Fr = \frac{G^2}{9.8 \rho_l^2 d}$ $\Delta H = \frac{Cp(T_{sat} - T_m)}{h_f}$ $L_e = \frac{m[Cp(T_{sat} - T_m) + h_{fg} x_{CHF}]}{\pi dq_{CHF}}$
Styrikovich [12]	Uniform Heat Flux; Water; Vertical	$d = 10$; $D = 136$ $L = 1709$; $p = 55$	13.6	$P = 98.1\text{--}176$; $G = 500\text{--}1500$; $q'' = 100\text{--}1500$	Effect of flow direction (Up flow and Down flow); Flow direction does not affect CHF; In high quality region, CHF in helical coil is much higher than straight vertical tube
Jayanti [8]	Uniform Heat Flux; Freon - 12; Vertical	$d = 20$; $t = 1$ $D = 133, 480, 880, 1800$	6.65– 90	$P = 11\text{--}28.1$; $G = 500\text{--}2000$; $q'' = 10\text{--}40$	$x_{tot} = \Delta x + x_1$ $x_1 = 0.164 P^{0.153} D^{-0.109} G^{0.277} Q^{-0.23}$ $\Delta x = 1.087 - 1.663 \times 10^{-3} P - 6.03 \times 10^{-4} D - 1.463 \times 10^{-2} G - 3.623 Q - 1.05 x_1 - 6.01 \times 10^{-2} Pr_g$ $Q = \frac{q''}{G h_{fg}}; 3 \times 10^{-5} < Q < 90 \times 10^{-5}$ $P = \frac{\rho_l}{\rho_g}; 5 < P < 20$ $D = \frac{D_c}{d}; 6 < D < 80$ $G = \varphi_{CHF}; 2.5 < G < 11$ $\varphi_{CHF} = \left[\frac{Gd}{\mu_l} \left(\frac{\mu_l^2}{\sigma d \rho_l} \right) \left(\frac{\mu_l}{\mu_g} \right)^{\frac{1}{5}} \right]$
Ma [13]	Uniform Heat Flux; Water; Horizontal	$d = 10\text{--}11$ $D = 132, 256$ $L = 2413, 7560$ $t = 2$	13.2, 23.3	$P = 10\text{--}40$; $G = 280\text{--}850$; $q'' = 100\text{--}1200$	$q''_{CHF} = A - Bx_{CHF}$ $G \leq 440$ $G > 440$ $A = 42.3(d/D)^{0.11} (G - 179.2)$ $A = 25.7(d/D)^{0.11} (G - 16.2)$ $B = 38.8(d/D)^{0.06} (G - 184.7)$ $B = 21.6(d/D)^{0.06} (G + 11.3)$
Kaji [14]	Uniform Heat Flux; R113; Vertical	$d = 10$ $t = 0.19$ $D = 165, 320$	16.5– 32	$P = 3.9$; $G = 305\text{--}1560$; $q'' = 30\text{--}200$	$\frac{q''_{CHF}}{G h_{fg}} = 0.34 \left(\frac{\sigma \rho_l}{G^2 L} \right)^{0.043} \left(\frac{1}{L_n/d} \right) \left\{ 1 + 3.5 \left(\frac{d}{D} \right)^{1.2} \right\}$
Chung [15]	Uniform Heat Flux; R113; Vertical	$d = 10.9$; $t = 0.89$ $D = 215, 485$ $L = 4100$	20, 44	$P = 3\text{--}7$; $G = 300\text{--}500$; $q'' = 36\text{--}80$	Influence of coil diameter, pressure and mass flux on dryout patterns
Matsui [16]	Oil Heated Water, Nitrogen; Vertical	$d = 10$; $t = 1$ $D = 200$; $N = 2.5$	20	$Toil = 200\text{--}205$ °C	Studied the Phenomena of DNB in low quality region
Ami [17]	Uniform Heat Flux; Liquid Nitrogen; Vertical	$D = 100$; $L = 3150$ $d = 4$; $t = 1$	25	$P = 3$; $G = 200\text{--}710$; $q'' = 0\text{--}60$	Macbeth's correlation for straight tube and/or Kaji's correlation for helically coiled tube Works well to Predict CHF in helical coil
Chen [18]	R134a Horizontal	$\frac{q''_{CHF}}{G h_{fg}} = f \left(\varphi_{CHF}, \frac{\Delta h}{h_{fg}}, \frac{\rho_l}{\rho_g}, \frac{L}{D}, \frac{L^n}{D^n} \right); \varphi_{CHF} = \left[\left(\frac{GD_n}{\mu_l (\sigma h_{fg})^{n-1}} \right)^a \left(\frac{\mu_l^2 (\sigma h_{fg})^{n-1}}{\sigma D^n \rho_l} \right) \left(\frac{\mu_l}{\mu_g} \right)^c \right]$ $n = 1.62, a = 1, b = 0.38, c = 0.14$		$D_n = L^n + \left(\frac{D}{p} \times p \right)^n + \left(\frac{D}{L} \times p \right)^n$	
Chen [19]	Uniform Heat Flux; R134a; Horizontal	$d = 3.8\text{--}11$ $D = 135\text{--}370$ $p = 40\text{--}105$ $L = 853, 7541$	17.8– 78.9	$P = 3\text{--}11$; $G = 60\text{--}480$; $x_1 = -0.32$ to 0.36 ; $q'' = 6\text{--}90$	Dryout - $Bo = 8.071 \times 10^{-8} \times Re_l^{2.12} Dn^{-2.5} N_d^{0.18} x_1^{-0.78}$ DNB - $Bo = 3.154 \times 10^{-7} \times Re_l^{2.45} Dn^{-2.1} N_d^{0.46} x_0^{-0.25}$ $N_d = \rho_l / \rho_g$ $Dn = Re_l (d/D)^{0.5}$ $Re_l = \frac{Gd}{\mu_l}$
Chen [20]	Uniform Heat Flux; R134a; Horizontal	$d = 8.4$; $t = 0.8$ $D = 300$; $p = 75$ $L = 1890$	35.7	$P = 3\text{--}9.5$; $G = 60\text{--}500$; $q'' = 7\text{--}50$; $x_1 = -0.36$ to 0.35	$Bo = 1.135 \times 10^{-7} \times Re_l^{2.32} Dn^{-2.5} N_d^{0.19} x_1^{-0.58}$
Tan [21]	Uniform Heat Flux; R134a; Horizontal	Fluid to Fluid Modeling		$P = 2\text{--}5$; $G = 50\text{--}1500$	Ahmed and Katto Correlation used for Helical Coiled Tubes for $\varphi_{CHF-Ahmed} > 15$ $\varphi_{CHF-katto} > 50$
Chung [22]	Uniform Heat Flux; Water; Vertical	$d = 12$; $t = 2.5$ $D = 577, 937, 1297$ $L = 2050$	48–108	$P = 10\text{--}60$; $G = 176\text{--}530$	CHF occur at quality nearly 1.0
Hwang [6]	Uniform Heat Flux; Water; Vertical	$d = 12$; $t = 2.5$; $D = 606, 977$; $L = 2000$	50.5, 81.4	$G = 88.4\text{--}530.5$; $P = 10\text{--}60$; $q'' = 30\text{--}1145.3$	In most of the data, CHF occur at quality greater than 0.9; Modified Berthoud and Jayanti Dryout Map
Santini [23]	Uniform Heat Flux; Water; Vertical	$d = 12.5$; $D = 1000$; $p = 790$; $L = 32,000$	80	$P = 10\text{--}61$; $G = 199\text{--}810$; $q'' = 40\text{--}209$	$x_1 = (0.44 - 0.0006P)G^{0.114}$; P in bar

the helical coil and tube diameter of coil is measured before bending the straight tube. The heated length of helical coil is average of inner side and outer side length. The uncertainty in the heated

length is obtained by repeatedly measuring the values. The change in the tube diameter because of ovality due to bending of tube is accounted in the uncertainty of the tube diameter.

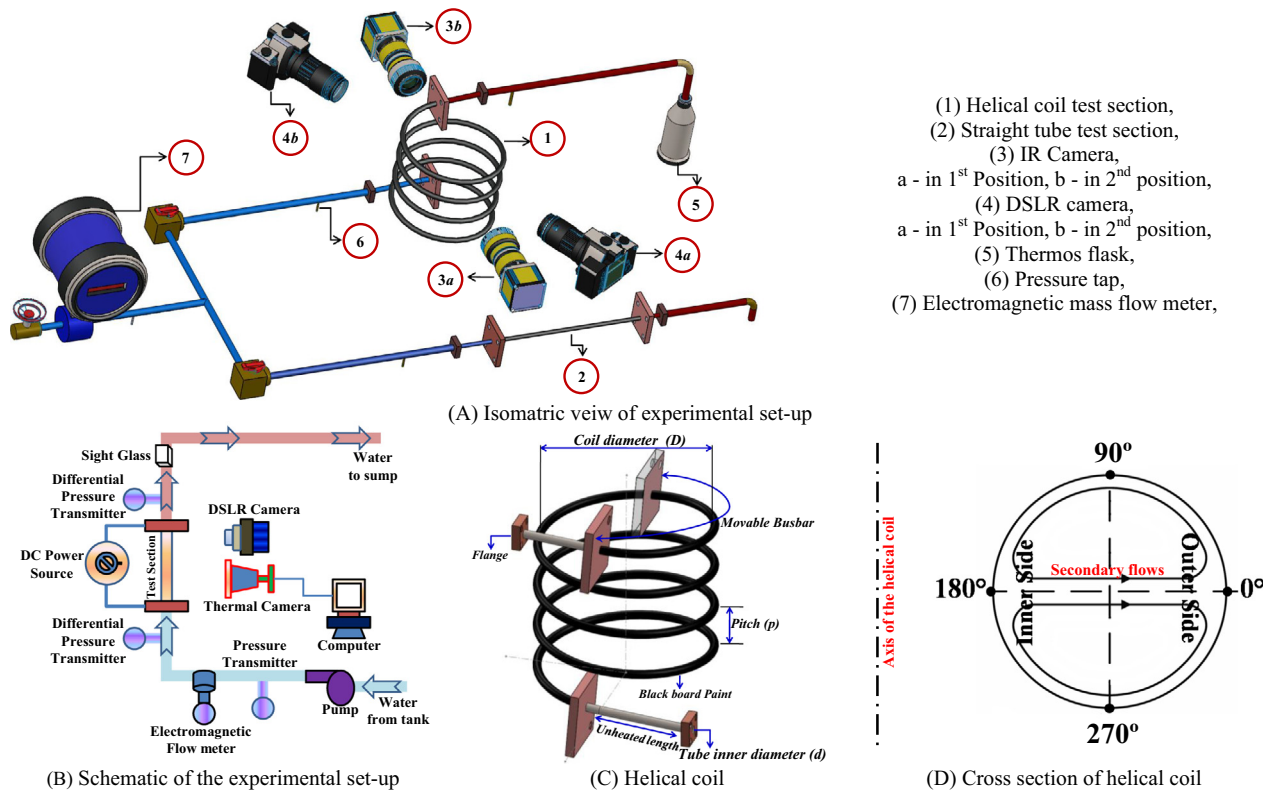


Fig. 3. The experimental set-up.

Table 4
 Geometric details of test sections used for CHF analysis.

Coil No.	Tube characteristics		Coil characteristics		Curvature ratio	System pressure	Mass flux	Quality	No of CHF data points
	Inner diameter d (mm)	Outer diameter d_o (mm)	Mean diameter D (mm)	Pitch p (mm)	D/d	P (bar)	G (kg/m ² s)	x	
1	6	8	162	50	27	1.2–6.3	228–1081	0.10–0.96	47
2	6	8	347	50	57.83	1.2–5.8	229–1241	0.11–0.78	29
3	8	9.4	137	50	17.13	1.2–3.8	192–949	0.03–0.78	18
4	8	9.4	383	50	47.88	1.2–5.0	166–507	0.14–0.95	22
5	9.7	11	140	50	14.43	1.1–4.0	92–830	0.00–0.97	31
6	10	11	301	50	30.1	1.1–3.5	188–442	0.13–0.92	9

Table 5
 Experimental uncertainties.

Parameter	Relative uncertainty (%)	Parameter	Relative uncertainty (%)
Pipe diameter	3.0	Mass flow rate	1.8
Pipe length	0.5	Heat flux	3.3
Coil diameter	1.0	Mass flux	3.5
Current	0.5	Quality	4.1
Voltage	1.0	Saturation temperature	1.5
Inlet temperature	1.3	Boiling number	4.6
System pressure	1.5		

3. Results and discussion

Experiments are conducted on critical heat flux with water for six helical coil test section in vertical orientation. Critical heat flux is detected using the “Red Hot” method.

- Heat flux at which a temperature rises sufficient to burst the pressurized heated tube or to heat part of the tube to red (suited both for dryout and DNB crisis types).

The results obtained for critical heat flux in all test sections are presented in [Appendix A](#). Following details are specified along with CHF data, the heated length of test section, system pressure which gives saturation temperature, mass flux, quality, critical heat flux and CHF ratio. CHF ratio (normalized CHF) is defined as the ratio of experimental CHF in helical coil and CHF in straight vertical tube as given by Eq. (6). The CHF in a straight vertical tube is calculated from Groeneveld et al. [25] look-up table. CHF from the look-up table is calculated using direct substitution method i.e. system pressure, mass flux, and quality remain same.

$$CHF \text{ Ratio} = \frac{CHF_{Coil}}{CHF_{Look-up \text{ Table}}} \quad (6)$$

3.1. Description of red hot method use to predict critical heat flux

The red hot method is used to detect CHF in the present work. In the red hot method, the value of heat flux when the portion of

helical coil jumps to a high temperature and becomes red is considered as critical heat flux. The heat flux supplied to the test section is increased in small steps. Once the temperature rises sufficiently high, the sequence of burnout occurs at almost constant heat flux. With the passage of time the portion of tube became more and more red until the tube gets break. The sequence of events occurs during a typical CHF experiment is shown in Fig. 4. Fig. 4 shows the state of helical coil test section with increase of time at constant heat flux. The inner side and outer side of helical coil are recorded at 150° opposite direction to see the bottom portion of the helical coil. As accepted, the burnout starts from inner side of helical coil. Initially because of high temperature, the black body paint gets burn. Unlike the straight tube, burnout in helical coil spread in a large portion. The inner side become red while the outer side is still at low temperature. With the time, red fraction spread on top and bottom side. The resistivity of helical coil increase with increase in temperature which slightly decreases current and hence the heat flux. Before bursting the tube, burnout spread on complete circumference of coil. After bursting, the fraction of burnout decreases with contact of water.

3.2. Wall temperature distribution at the time of critical heat flux

Hardik et al. [26] describe the wall temperature distribution in single phase flow inside helical coil. The secondary flows affect the temperature distribution of helical coil. The secondary flows make the circumferential wall temperature profile non-uniform on inner side and on outer side. The centrifugal force and density difference between hot and cold fluid makes the high temperature on inner side and low temperature on outer side. In a two phase flow, the density difference between water and vapor is very high. High density difference makes the temperature profile more non-uniform. Fig. 5 shows the temperature distribution along the axial length of coil on inner side and outer side of helical coil at 0° and 180° respectively. Temperature profile with increase of heat flux and exit quality is drawn. The temperature is captured till the starting of CHF. The temperature profile drawn from low heat flux value to the heat flux when burn out starts. The value of heat flux when burnout starts is considered at critical heat flux. With the increase of heat flux, temperature on the inner side increases at a higher rate than that of outer side. As the quality of vapor increases, more

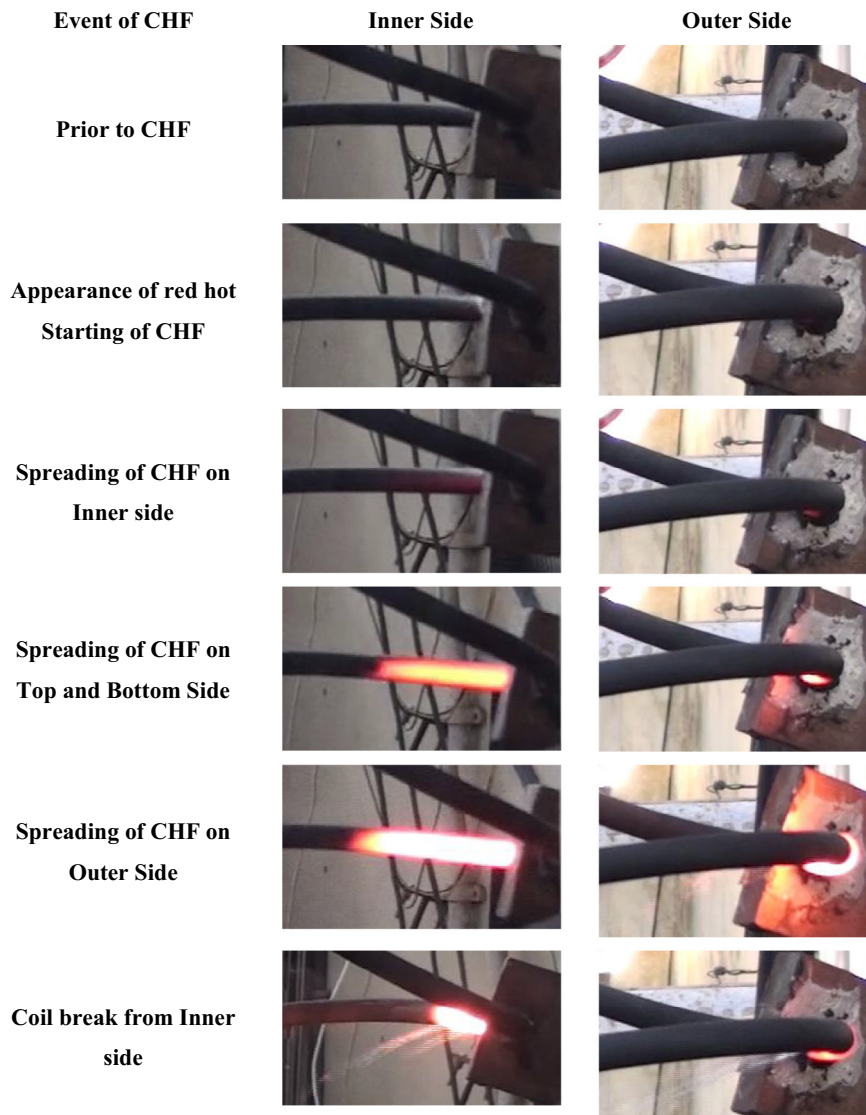


Fig. 4. Occurrence of critical heat flux for Coil - 3; CHF = 1759 kW/m²; x = 0.22.

vapour moves toward inner side. The secondary flows move water droplets from outer surface to inner surface. Less heat transfer through vapour increases the inner side wall temperature at higher rate compared to that of the outer side. Burnout starts from inner side. Once the inner side temperature reaches to the burnout point, it jumps sharply to high temperature.

The wall temperature on inner side and outer side of helical coil near the end of the test section (at axial location $L/d = 80$) at different heat flux is compared. The wall temperature variation with the heat flux curve is shown in Fig. 6. The non-uniformity in wall temperature around the circumference of helical coil decreases in the subcooled region with the decrease in subcooling. Non-uniformity is less near zero quality. As the quality increases, the non-uniformity between inner side and outer side wall temperature increases. Near the critical heat flux wall, temperature difference on inner side and outer side becomes very large.

3.3. Effect of tube diameter on critical heat flux

Fig. 7(A) shows the effect of tube diameter on critical heat flux. CHF data of two helical coils with tube diameter 8 mm and 9.7 mm compare for different quality. The coil diameter of both the helical coils is kept same to signify the effect of tube diameter on critical heat flux. The literature on CHF in straight tube concluded the decrease in CHF value with increase in tube diameter. Effect of tube diameter on CHF in helical coils is similar to straight tubes. CHF value in helical coil decreases with increase in tube diameter. The plot of non-dimensional CHF with quality is shown in Fig. 7 (B). The tube diameter effect is included in straight tube CHF calculation. The tube diameter effect on helical coil CHF is same as on straight tube CHF. The non-dimensional CHF values of both the tubes merge well.

3.4. Effect of coil diameter and curvature ratio on critical heat flux

The critical heat flux data at different quality for two different coil diameters with same tube diameter is studied. Critical heat flux values for 8 mm tube diameter and 137 mm and 383 mm coil diameters is compared in Fig. 8(A). Increase in coil diameter decreases the critical heat flux. This is in line with the literature as describe in Table 1. For the same tube diameter, as the coil diameter and hence diameter ratio increases, CHF values decreases. Fig. 8(B) shows the CHF values in helical coils with different tube diameter ($d = 6$ mm and 8 mm) and different coil diam-

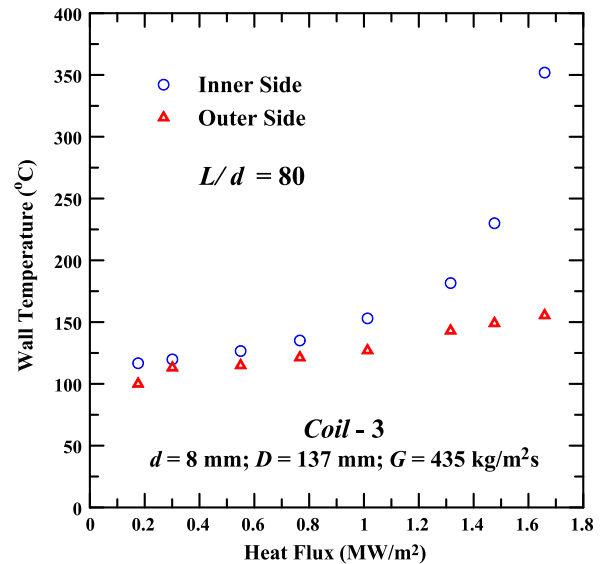


Fig. 6. Variation of the wall temperature with the heat flux.

eter ($D = 347$ mm and 140 mm). The critical heat flux in a coil with 8 mm tube diameter and diameter ratio (D/d) 17 is lower than the CHF values in a coil with 6 mm tube diameter and 58 diameter ratio. The effect of the tube diameter is larger on CHF value compared to the coil diameter. Hence, for different tube diameters, CHF in helical coil cannot be decided on the basis of the diameter ratio.

Fig. 8(C) and (D) shows the ratio of CHF in helical coil with vertical straight tube CHF at different quality. The effect of coil diameter is considered in normalized CHF. The CHF values of helical coils with same tube diameter and two different coil diameters are overlapping for normalized CHF variation with the quality. The CHF values of helical coils with different tube diameters and different coil diameters are also overlapping on normalized CHF variation with the quality graph. The change in coil diameter may changes the pressure drop and hence the system pressure. The effect of the system pressure is included in calculation of straight tube CHF. Hence, the normalized CHF considers the effect of change in coil diameter as well as change in tube diameter on CHF of a helical coil.

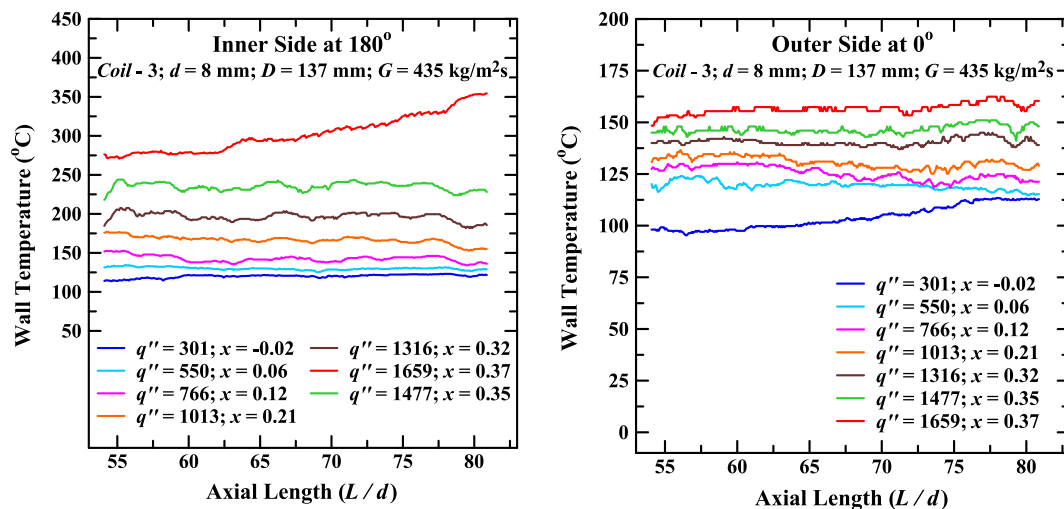


Fig. 5. Wall temperature distribution in helical coils.

3.5. Effect of mass flux on critical heat flux

The literature reported increase in mass flux increases critical heat flux. Experimental CHF data of all the helical coils with

increase in mass flux are shown in Fig. 9. Increase in the mass flux increases the CHF. Increase in the mass flux increases the convection boiling heat transfer which delays the burnout. The CHF variation with the mass flux data of all the coils with different tube

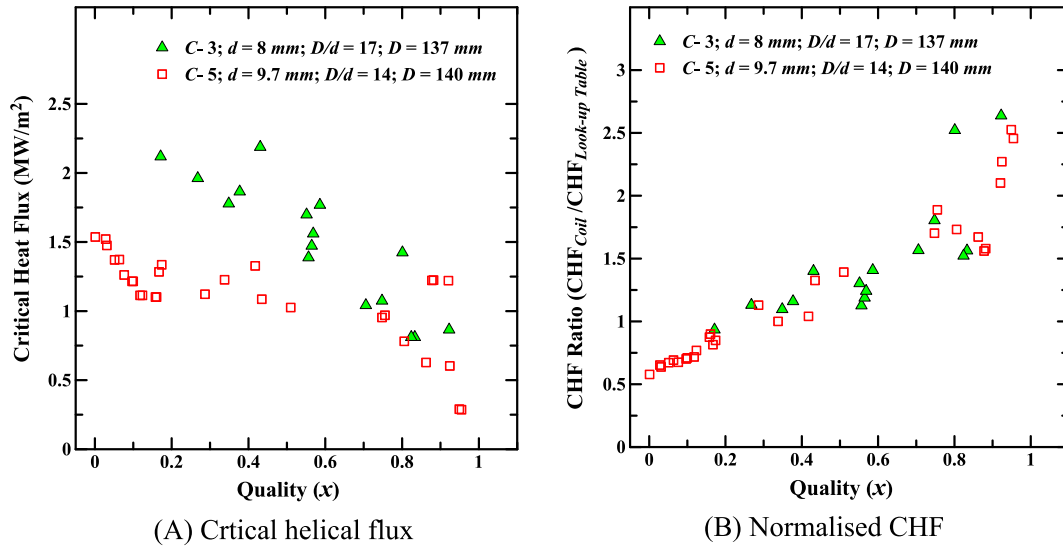


Fig. 7. Effect of tube diameter on critical heat flux.

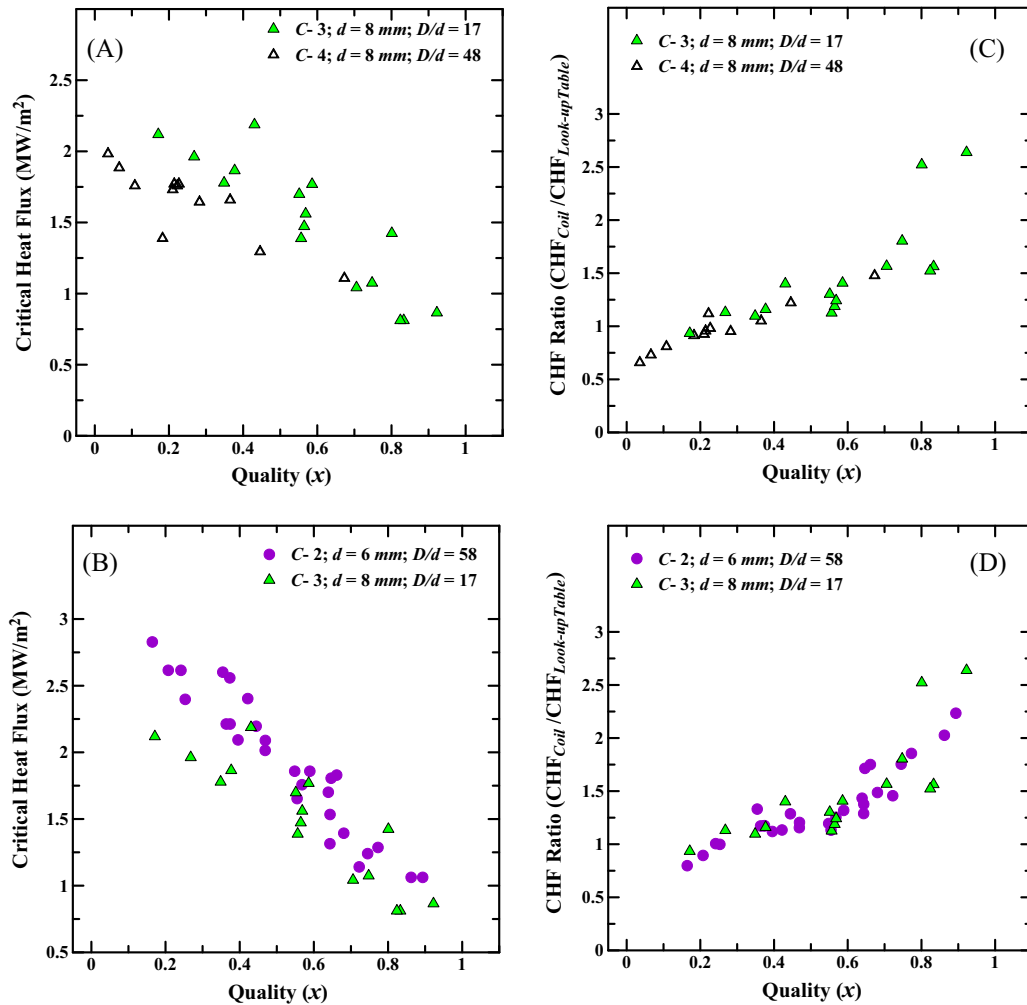


Fig. 8. Effect of curvature ratio on critical heat flux.

diameters and different coil diameters are nearly overlapping. The effect of mass flow in different tube diameter on critical heat flux is well accounted with the mass flux. With constant mass flux, critical heat flux values remain uniform for different tube diameter and coil diameter. Data presented in the normalized CHF variation with the mass flux is not providing much information. In a low mass flux region, CHF ratio varies from 1.3 to 2.5 for constant mass flux.

3.6. Effect of quality on critical heat flux

The quality reported in the present study is the thermodynamic equilibrium quality at exit of heated length of the test section. The tabular data of CHF from Appendix A is shown in graphical form as critical heat flux against quality. The effect of increase of quality on CHF is shown in Fig. 10 for all the helical coils tested in present study. In all the helical test sections, as the quality increases, the value of critical heat flux decreases.

for all the helical coils with different tube diameters and different coil diameters nearly collapse into a single curve. The normalized CHF values increase with increase in the quality. The critical heat flux values for identical mass flux are almost same. For the same mass flux in a small diameter tube, the mass flow rate is high. At a specific quality, as the flow rate increases, the heat flux increases. Hence, for the same quality, critical heat flux decreases as the tube diameter increases.

3.7. Effect of tube diameter, coil diameter and mass flux on critical quality

The quality at a point of burnout is considered as a critical quality. In all the experiments, burnout (Dryout) occurs near the end of the test section. Hence in the present study, the thermodynamic equilibrium quality at exit of the test section is nearly same as critical quality. Fig. 11 shows the effect of mass flux on critical quality.

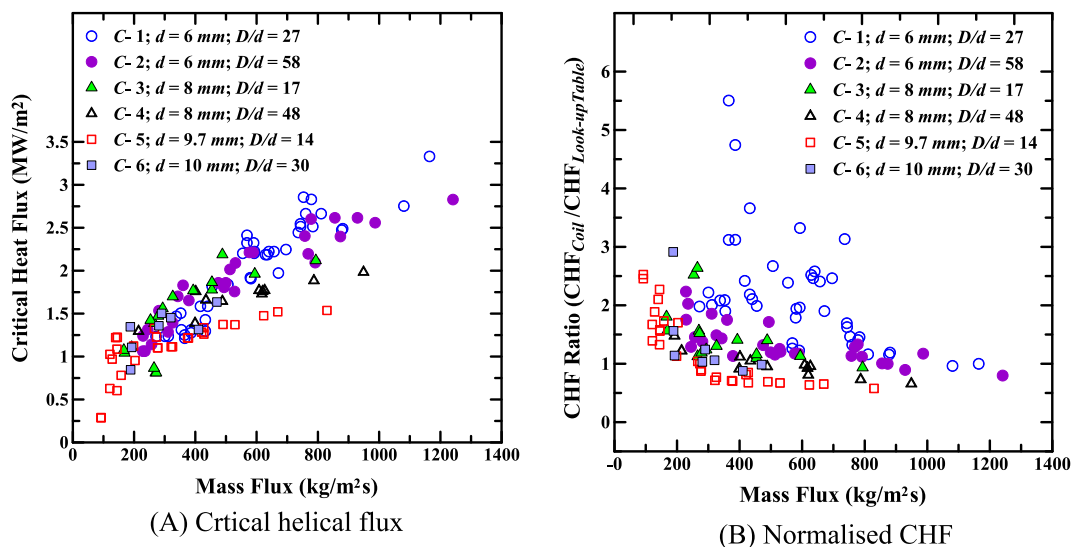


Fig. 9. Effect of mass flux on CHF for different tube diameters and curvature ratios.

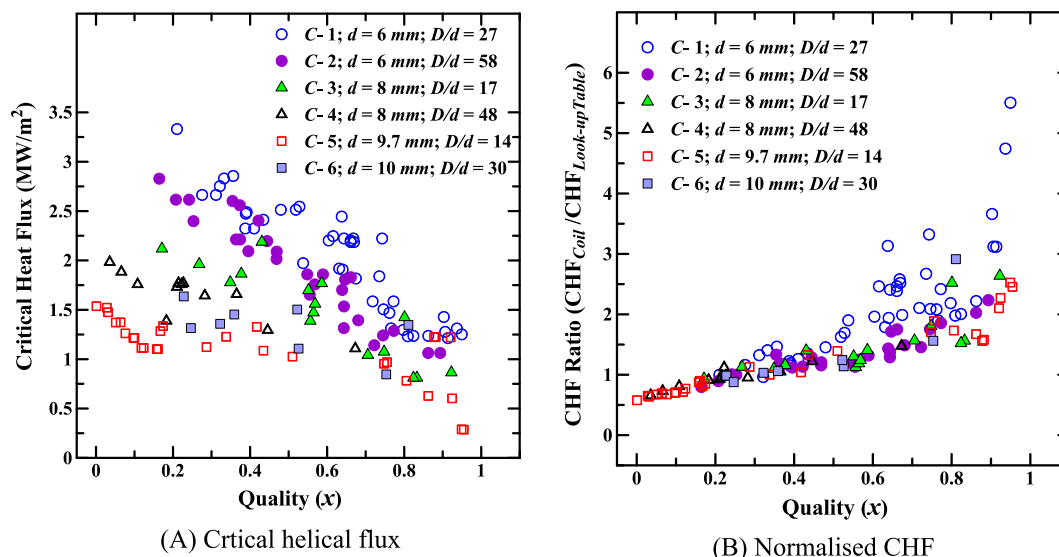


Fig. 10. Effect of tube diameter and curvature ratio on CHF for different quality.

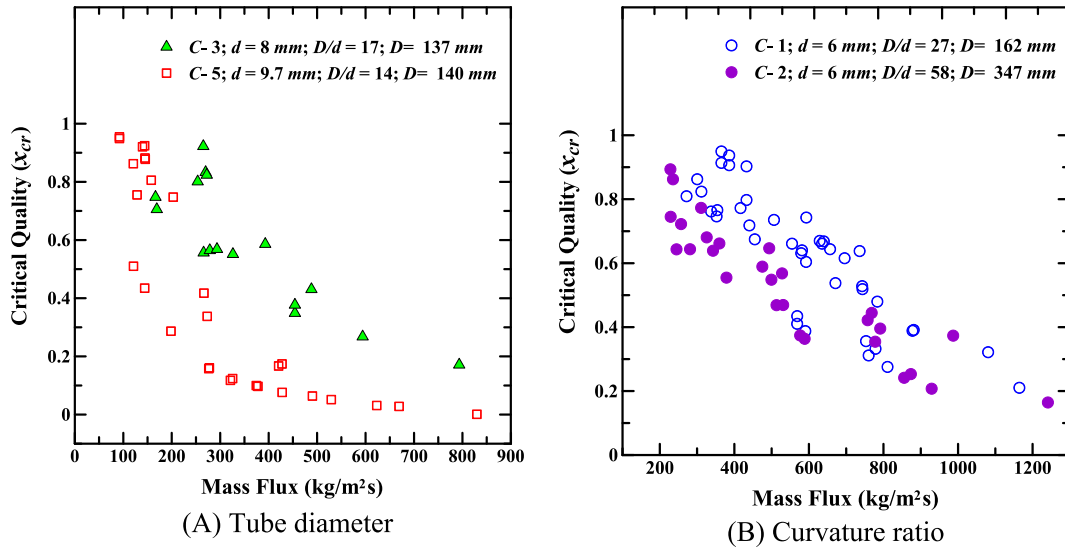


Fig. 11. Effect of tube diameter, curvature ratio and mass flux on critical quality.

Opposite to CHF variation pattern, the critical quality decreases with increase in the mass flux. The effect of change in tube diameter and curvature ratio is shown in Fig. 11(A) and (B) respectively. Similar to CHF variation pattern, the critical quality decreases with the increase in the tube diameter and coil diameter. The critical heat flux data along with the corresponding quality shows effect of tube diameter and coil diameter. The critical heat flux data variation with the mass flux is overlapping for different tube diameters and different coil diameters. The critical quality variation with the mass flux shows the effect of tube diameter and coil diameter. The tube diameter and coil diameter affect the critical quality for constant mass flux. Hence, the change in tube diameter and coil diameter mainly change the critical quality.

3.8. Experimental data comparison with literature

Berthoud and Jayanti [5] collected experimental CHF data from the literature and derived a chart to divide CHF in different region. The chart is shown in Fig. 1. The present experimental CHF data are superimposed on that chart. Fig. 12 shows the data with properties

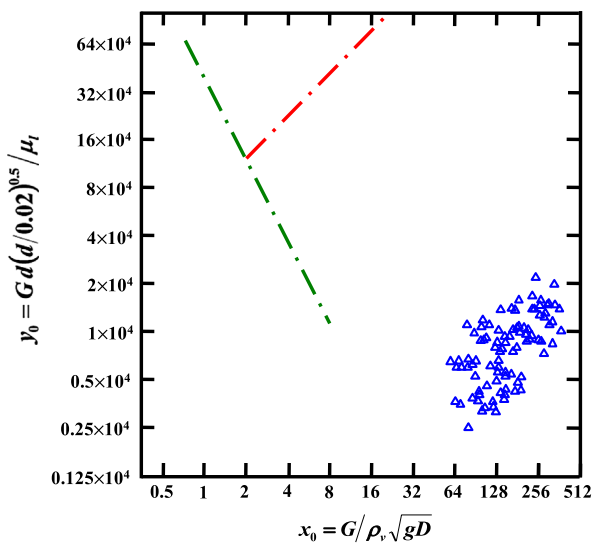


Fig. 12. CHF Data superimposed on Berthoud chart.

taken at present experimental CHF condition are superimposed on Berthoud and Jayanti [5] chart. The chart is limited for X-coordinate up to 30 and Y-coordinate from 20,000 to 640,000. The present data range is quite far from the chart range. The present data is ranged for X-coordinate from 60 to 512 and Y-coordinate from 2500 to 25,000. Hence, there is less probability that the correlation derived by Berthoud and Jayanti [5] and the other literature [8,11,12] referred by Berthoud and Jayanti [5] work for present data prediction. Similar to this, the data measured by Hwang et al. [6] are far from present data range.

The CHF for experimental data range is predicted using the helical coil correlations available in literature. The predicted CHF values from correlations are compared with present experimental values. CHF calculated using Merilo [27] correlation (Eq. (7)) for straight horizontal tube is compared with the experimental values. Comparison is shown in Fig. 13. Available CHF correlations for helical coils are not able to predict the CHF data for low pressure water system. Jensen and Bergles [7] derived correlation for R113 in helical coils. This correlation overpredicts CHF in a low quality region and underpredicts in a high quality region. The scatter in the data predicted using Jensen and Bergles [7] is large. Merilo [27] correlation overpredicts CHF in comparison with the present experimental values. As expected, correlations available in the literature are derived for high system pressure and for low density ratio (ρ_l/ρ_g). Hence, they are not able to predict the CHF for experimental data which, measured in low system pressure and high density ratio. It is necessary to derive an equation to predict CHF data for the present range. Deviation in a data predicted using Merilo [27] correlation is large but the scatter in a data is less. A constant in Merilo [27] correlation is changed from 525 to 110. Comparison of experimental CHF with modified Merilo [27] correlation is shown in Fig. 14(A). This modified Merilo [27] correlation predicts 70% of experimental data within 30% deviation. Fig. 14(B) compares the experimental CHF with modified Merilo [27] correlation for different quality. The correlation is deviating large in a low quality region. The correlation predicts the data reasonably well in a high quality region ($x > 0.20$). Baburajan et al. [28] performed experiments on CHF in low pressure straight horizontal tube. Experimental data of the study predicted well using the modified Merilo [27] correlation with the value of constant is 25.97 instead of 525. Compare to straight horizontal tube, the CHF in helical coil is almost 4 times higher in a high quality region.

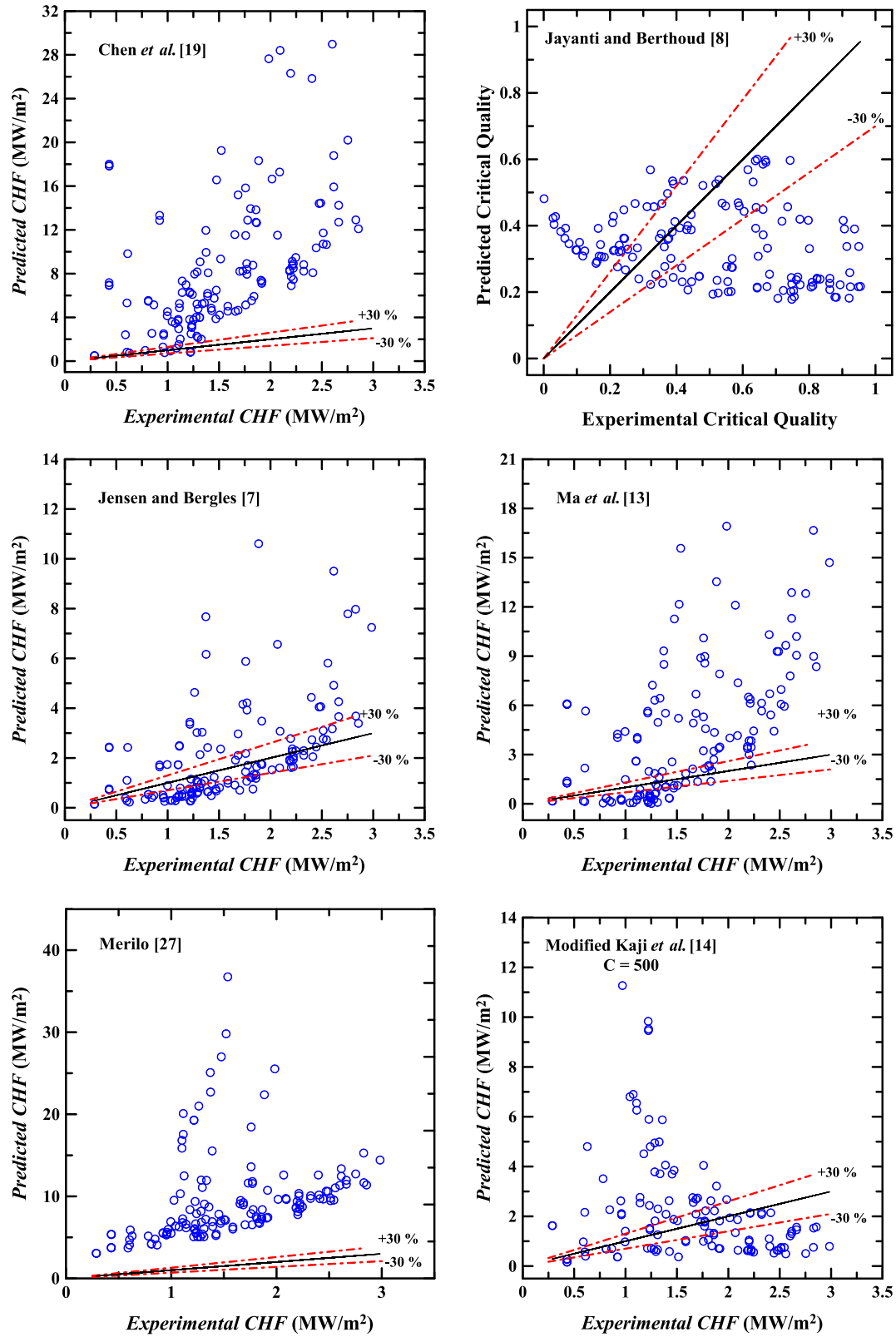


Fig. 13. Comparison of available correlations with experimental CHF Data.

$$\frac{q''}{Gh_{fg}} = CRe_l^{-0.34} \{Z^3 B_o\}^{0.358} \left(\frac{\mu_l}{\mu_g}\right)^{-2.18} \left(\frac{L_h}{d}\right)^{-0.511} \left(\frac{\rho_l}{\rho_g} - 1\right)^{1.27} (1 - x_i)^{1.64} \quad (7)$$

$$Re_l = \frac{Gd}{\mu_l}; \quad Z = \frac{\mu_l}{(\sigma d \rho_l)^{0.5}}; \quad B_o = \frac{(\rho_l - \rho_g)gd^2}{\sigma}; \quad x_i = \frac{Cp(T_i - T_{sat})}{h_{fg}}$$

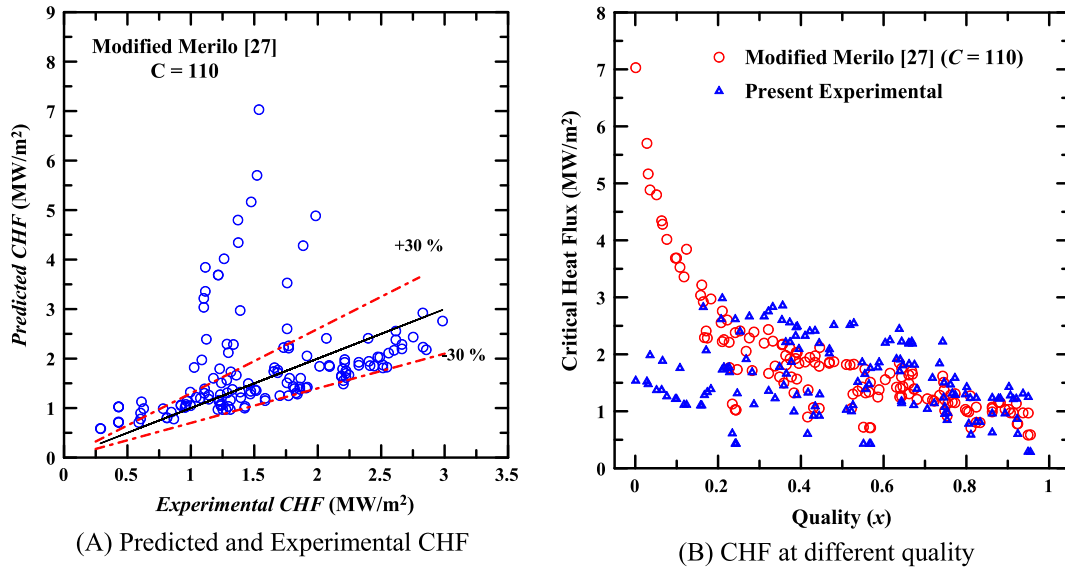


Fig. 14. Comparison of modified Merilo [27] correlation with experimental CHF data.

$C = 575$ – Suggested by Merilo [27] for high pressure horizontal tubes (8)

$C = 25.79$ – Suggested by Baburajan et al. [28] for low pressure horizontal tubes (9)

$C = 110$ – Suggested in Present work for low pressure helical coils (10)

The experimental CHF data is normalized with Groeneveld et al. [25] look-up table data. The normalized CHF data is presented against quality in Fig. 15. The scatter seen in critical heat flux for different tube diameters is merged in Fig. 15. The variation of normalized CHF with the quality follows a linear curve. Regression analysis of measured CHF data is given by Eq. (11). The experimental results of the present study are compared with present correlation given by Eq. (11). The CHF in helical coil is linear function of quality. The correlation gives simple method to calculate CHF in helical coil from look-up table CHF data. The correlation predicted

90% of experimental data within 25% deviation. Root mean square (RMS) deviation of the correlation with experimental data is 17% and average deviation is -1.6% .

$$\frac{CHF_{Coil}}{CHF_{Look-up Table}} = 1.637x + 0.568 \quad (11)$$

Fig. 16 compares the helical coils CHF with look-up table CHF data. The normalized CHF is increased with increase in quality. This is may be because of the effect of gravity and redeposition of liquid film due to secondary flow. Ideally, vertical helical coil with small curvature behaves as a horizontal tube. In the low quality region, the effect of gravity is higher and the effect of redeposition is lower. Hence, in the low quality region, CHF is lower than vertical tube because of gravity effect but higher than horizontal tube due to redeposition effect. As the quality increases, the effect of redeposition increases and the effect of gravity decreases. Hence, CHF in a

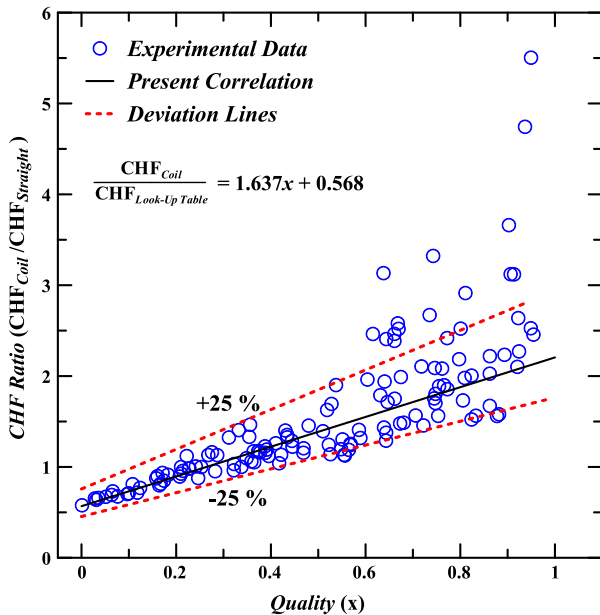


Fig. 15. Comparison of experimental CHF with present correlation.

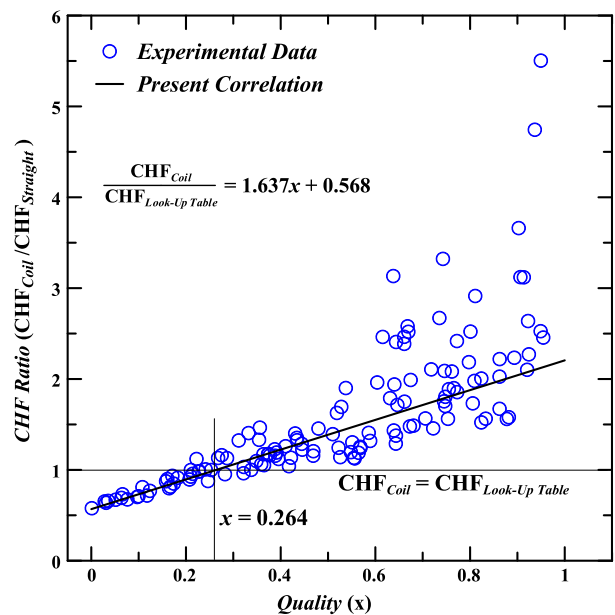


Fig. 16. Comparison of helical coil CHF with straight tube CHF data.

helical coil is higher than the vertical tube in a high quality region and normalized CHF increases with increase in quality. At higher quality, the CHF value in a straight tube is lower than the helical coil. Hence the scatter in normalized CHF is more in high quality region. The helical coil CHF is nearly same as straight tube CHF at quality around 0.26. At zero quality, helical coil CHF is 40% less than straight vertical tube CHF value. At quality > 0.6, helical coil CHF value is 1.5–3 times that of straight vertical tube CHF.

4. Conclusions

Experiments are performed to study critical heat flux in helical coils at low pressure. Six helical coil test section with three tube diameters 6 mm, 8 mm and 10 mm and D/d ratio from 14 to 58 are tested. The effect of mass flux, quality, coil diameter and tube diameter on critical heat flux and critical quality are analysed. Following are the conclusions that may be drawn from the present study.

- The CHF in helical coil is affected because of secondary flows.
- The increase in coil diameter and tube diameter decreases the CHF. Curvature ratio does not properly represent the change in CHF.
- In saturated boiling region, the CHF decreases with the increase in quality and increases with increase in mass flux. The critical quality decreases with increase in mass flux.
- In the low quality region, CHF in a helical coil is lower than the straight vertical tube CHF. However, for high quality region, CHF in a helical coil is higher than the straight vertical tube CHF.

- The CHF in helical coil simply correlates with straight tube CHF data and quality.
- A simple correlation is developed to estimate the CHF in a helical coil.

5. Contributions of the present work

The work presents a comprehensive literature review and status of research on critical heat flux in helical coils. Correlations used to predict critical heat are identified. Local wall distribution along the axial length of helical coils aids in identifying hot spots in a helical coil. Parametric study shows the comparison of critical heat flux with tube diameter, coil diameter, quality and mass flux. Simple correlation derived in the present study helps to predict the critical heat flux in low pressure system. Helical data comparison with straight vertical tube data is useful to select the geometry in high heat flux application.

Acknowledgements

Authors hereby acknowledge the financial support given by Ministry of Defence (R and D). Authors wish to acknowledge the support given by Captain Binduraj from Ministry of Defence (R and D), Shri K.N. Vyas and Shri Joe Mohan from Bhabha Atomic Research Centre and Dr. P.K. Baburajan from Atomic Energy Regulatory Board. Authors are grateful to Mr. Rahul Shirsat for his assistance in building the experimental setup and fabrication of test sections.

Appendix A. Experimental critical heat flux data

Sr. No.	Coil No.	Heated length (mm)	Mass flux (kg/m ² s)	System pressure (bar)	Exit quality	Heat supply (W)	Critical heat flux (kW/m ²)	CHF ratio
1	Coil_1	503.5	568	4.33	0.410	22,054	2324	1.260
2	Coil_1	503.5	569	4.38	0.434	22,896	2412	1.352
3	Coil_1	503.5	576	4.11	0.374	21,000	2213	1.175
4	Coil_1	503.5	589	3.97	0.363	21,000	2213	1.172
5	Coil_1	503.5	590	4.25	0.387	22,054	2324	1.227
6	Coil_1	503.5	753	4.91	0.356	27,104	2856	1.465
7	Coil_1	503.5	760	4.83	0.311	25,281	2664	1.324
8	Coil_1	503.5	779	4.83	0.332	26,862	2830	1.405
9	Coil_1	503.5	811	4.91	0.276	25,281	2664	1.161
10	Coil_1	503.5	855	4.75	0.242	24,824	2616	1.005
11	Coil_1	503.5	929	4.45	0.207	24,824	2616	0.894
12	Coil_1	722.7	272	2.64	0.809	16,783	1232	1.979
13	Coil_1	722.7	338	3.22	0.762	20,008	1469	2.082
14	Coil_1	722.7	352	3.04	0.746	20,501	1505	2.089
15	Coil_1	722.7	455	3.94	0.675	24,752	1817	1.989
16	Coil_1	722.7	555	4.75	0.661	30,000	2202	2.387
17	Coil_1	722.7	592	4.70	0.604	30,000	2202	1.962
18	Coil_1	722.7	743	5.60	0.528	34,668	2545	1.693
19	Coil_1	722.7	744	5.48	0.519	34,240	2513	1.626
20	Coil_1	722.7	784	5.42	0.480	34,240	2513	1.455
21	Coil_1	722.7	878	5.57	0.389	33,696	2474	1.155
22	Coil_1	722.7	881	5.48	0.391	33,904	2489	1.191
23	Coil_1	722.7	1081	5.96	0.322	37,496	2752	0.963
24	Coil_1	845	228	2.69	0.893	15,300	961	2.020
25	Coil_1	845	235	2.62	0.862	15,300	961	1.832
26	Coil_1	845	300	3.47	0.862	19,686	1236	2.219
27	Coil_1	845	312	3.47	0.824	19,686	1236	2.005

Experimental critical heat flux data (continued)

Sr. No.	Coil No.	Heated length (mm)	Mass flux (kg/m ² s)	System pressure (bar)	Exit quality	Heat supply (W)	Critical heat flux (kW/m ²)	CHF ratio
28	Coil_1	855	354	3.47	0.766	21,141	1312	1.899
29	Coil_1	845	417	4.51	0.772	25,249	1585	2.418
30	Coil_1	845	440	4.47	0.718	25,249	1585	2.106
31	Coil_1	855	506	4.81	0.735	29,646	1839	2.671
32	Coil_1	845	579	5.52	0.631	30,550	1918	1.789
33	Coil_1	855	581	4.80	0.640	30,738	1907	1.940
34	Coil_1	845	593	6.31	0.743	35,400	2223	3.322
35	Coil_1	845	629	6.23	0.670	34,825	2186	2.519
36	Coil_1	845	635	6.16	0.661	34,825	2186	2.464
37	Coil_1	845	640	6.32	0.668	35,400	2223	2.580
38	Coil_1	845	657	6.36	0.644	35,400	2223	2.407
39	Coil_1	855	696	5.95	0.615	36,200	2246	2.463
40	Coil_1	855	736	6.30	0.638	39,395	2444	3.133
41	Coil_1	1100	365	3.96	0.914	25,150	1213	3.119
42	Coil_1	1100	365	3.96	0.949	25,942	1251	5.504
43	Coil_1	1100	386	4.30	0.906	26,476	1277	3.121
44	Coil_1	1100	386	4.30	0.937	27,181	1311	4.743
45	Coil_1	1100	433	4.48	0.798	26,873	1296	2.185
46	Coil_1	1100	433	4.48	0.903	29,600	1428	3.661
47	Coil_1	1100	671	4.91	0.538	31,416	1515	1.122
48	Coil_2	562	229	2.13	0.745	13,140	1240	1.754
49	Coil_2	562	245	2.08	0.643	12,478	1178	2.490
50	Coil_2	562	343	4.44	0.639	18,020	1701	1.433
51	Coil_2	562	359	4.44	0.599	18,020	1701	1.210
52	Coil_2	562	360	4.31	0.662	19,381	1830	1.751
53	Coil_2	562	365	2.71	0.539	16,482	1556	1.042
54	Coil_2	562	379	2.78	0.555	17,514	1653	1.132
55	Coil_2	562	513	3.45	0.469	21,344	2015	1.156
56	Coil_2	562	531	3.53	0.469	22,134	2089	1.205
57	Coil_2	562	778	4.33	0.354	27,560	2602	1.331
58	Coil_2	562	873	4.03	0.253	25,400	2398	0.999
59	Coil_2	562	1165	4.74	0.210	31,610	2984	0.895
60	Coil_2	562	1241	4.46	0.164	29,960	2828	0.798
61	Coil_2	677	257	2.94	0.722	14,559	1141	1.456
62	Coil_2	677	281	2.89	0.644	14,559	1141	2.236
63	Coil_2	677	326	3.61	0.681	17,784	1394	1.487
64	Coil_2	677	475	4.99	0.589	23,718	1859	1.319
65	Coil_2	677	499	4.99	0.548	23,718	1859	1.195
66	Coil_2	677	758	5.78	0.388	29,174	2286	1.024
67	Coil_2	677	758	5.78	0.422	30,669	2403	1.134
68	Coil_2	677	817	5.54	0.304	27,232	2134	0.888
69	Coil_2	764	311	2.77	0.773	18,530	1287	1.855
70	Coil_2	764	316	2.71	0.685	17,150	1191	1.333
71	Coil_2	764	493	3.90	0.647	26,010	1806	1.714
72	Coil_2	764	528	3.86	0.568	25,308	1757	1.254
73	Coil_2	764	768	4.77	0.444	31,625	2196	1.287
74	Coil_2	764	791	4.60	0.395	30,150	2094	1.120
75	Coil_2	764	987	5.36	0.373	36,848	2559	1.172
76	Coil_2	764	1143	5.08	0.249	34,010	2362	0.734
77	Coil_3	446.5	215	1.70	0.446	14,535	1295	1.222
78	Coil_3	446.5	399	1.87	0.183	15,584	1389	0.913
79	Coil_3	408.5	401	2.41	0.222	18,054	1758	1.120
80	Coil_3	446.5	620	2.21	0.108	19,740	1759	0.810
81	Coil_3	446.5	787	2.10	0.066	21,153	1885	0.731
82	Coil_3	446.5	949	2.08	0.036	22,261	1984	0.659
83	Coil_3	647.5	192	2.12	0.673	18,040	1109	1.478

(continued on next page)

Experimental critical heat flux data (continued)

Sr. No.	Coil No.	Heated length (mm)	Mass flux (kg/m ² s)	System pressure (bar)	Exit quality	Heat supply (W)	Critical heat flux (kW/m ²)	CHF ratio
84	Coil_3	647.5	203	2.11	0.627	18,040	1109	1.244
85	Coil_3	647.5	227	1.90	0.547	18,011	1107	1.069
86	Coil_3	647.5	382	2.74	0.363	23,375	1436	0.956
87	Coil_3	647.5	387	2.71	0.344	22,848	1404	0.930
88	Coil_3	647.5	397	2.31	0.253	19,125	1175	0.779
89	Coil_3	647.5	435	3.10	0.365	27,000	1659	1.052
90	Coil_3	647.5	594	3.56	0.268	31,164	1915	1.104
91	Coil_3	647.5	608	3.23	0.227	28,820	1771	0.981
92	Coil_3	647.5	617	3.25	0.210	28,171	1731	0.925
93	Coil_3	647.5	626	3.22	0.214	28,820	1771	0.957
94	Coil_3	647.5	793	3.76	0.171	33,660	2068	0.912
95	Coil_4	632	166	2.31	0.748	17,072	1075	1.804
96	Coil_4	632	169	2.29	0.706	16,556	1042	1.566
97	Coil_4	632	266	3.13	0.556	22,050	1388	1.126
98	Coil_4	632	278	3.22	0.565	23,391	1473	1.188
99	Coil_4	632	454	4.20	0.349	28,244	1778	1.097
100	Coil_4	632	454	4.20	0.377	29,625	1865	1.160
101	Coil_4	632	488	4.01	0.282	26,760	1685	0.976
102	Coil_4	782	254	4.00	0.801	27,987	1424	2.522
103	Coil_4	782	303	3.99	0.610	27,280	1388	1.184
104	Coil_4	782	393	5.02	0.586	34,765	1769	1.408
105	Coil_4	782	436	4.93	0.432	31,388	1597	1.011
106	Coil_4	1500.5	265	3.60	0.922	32,638	865	2.638
107	Coil_4	1500.5	270	3.51	0.833	30,600	811	1.563
108	Coil_4	1500.5	272	3.51	0.824	30,600	811	1.522
109	Coil_4	1500.5	478	4.36	0.444	34,752	922	1.291
110	Coil_4	1500.5	488	4.37	0.431	34,752	922	1.790
111	Coil_4	2294.5	293	3.27	0.569	24,789	430	2.357
112	Coil_4	2294.5	294	3.27	0.566	24,789	430	1.505
113	Coil_4	2294.5	300	3.27	0.551	24,789	430	2.026
114	Coil_4	2294.5	505	3.23	0.243	24,829	431	0.936
115	Coil_4	2294.5	505	3.24	0.243	24,829	431	0.936
116	Coil_4	2294.5	507	3.23	0.241	24,829	431	0.932
117	Coil_5	421.5	121	1.41	0.510	13,178	1026	1.393
118	Coil_5	421.5	144	1.48	0.435	13,951	1086	1.326
119	Coil_5	421.5	198	1.52	0.287	14,416	1122	1.129
120	Coil_5	421.5	276	1.48	0.158	14,151	1102	0.875
121	Coil_5	421.5	278	1.36	0.161	14,151	1102	0.899
122	Coil_5	421.5	321	1.47	0.118	14,310	1114	0.717
123	Coil_5	421.5	326	1.22	0.124	14,310	1114	0.769
124	Coil_5	421.5	374	1.49	0.100	15,615	1216	0.710
125	Coil_5	421.5	378	1.51	0.097	15,615	1216	0.701
126	Coil_5	421.5	428	1.50	0.076	16,207	1262	0.675
127	Coil_5	421.5	490	1.52	0.064	17,641	1373	0.692
128	Coil_5	421.5	529	1.42	0.051	17,612	1371	0.671
129	Coil_5	421.5	623	1.49	0.031	18,952	1475	0.638
130	Coil_5	421.5	669	1.38	0.028	19,542	1521	0.652
131	Coil_5	421.5	830	1.25	0.001	19,740	1537	0.578
132	Coil_5	617.5	273	2.33	0.338	23,089	1227	1.000
133	Coil_5	617.5	421	2.46	0.167	24,167	1284	0.815
134	Coil_5	617.5	428	2.51	0.174	25,129	1335	0.849
135	Coil_5	1059	121	1.72	0.862	20,264	628	1.672
136	Coil_5	1059	158	2.20	0.806	25,217	781	1.732
137	Coil_5	1059	203	2.68	0.748	30,775	954	1.702
138	Coil_5	2084	92	1.73	0.949	18,407	290	2.526
139	Coil_5	2084	92	1.71	0.955	18,170	286	2.456

Experimental critical heat flux data (continued)

Sr. No.	Coil No.	Heated length (mm)	Mass flux (kg/m ² s)	System pressure (bar)	Exit quality	Heat supply (W)	Critical heat flux (kW/m ²)	CHF ratio
140	Coil_5	658	128	2.09	0.755	19,450	970	1.888
141	Coil_5	658	266	3.17	0.417	26,618	1327	1.040
142	Coil_5	658	275	2.69	0.390	25,884	1283	1.010
143	Coil_5	658	470	3.95	0.228	33,790	1685	0.984
144	Coil_5	1406	140	2.24	0.921	24,960	1221	2.102
145	Coil_5	1406	144	2.30	0.924	25,845	603	2.271
146	Coil_5	1406	145	2.27	0.882	24,960	1224	1.579
147	Coil_5	1406	145	2.26	0.878	24,960	1222	1.560
148	Coil_6	678	192	2.50	0.526	23,575	1107	1.140
149	Coil_6	585	281	2.80	0.322	24,960	1358	1.032
150	Coil_6	678	320	3.34	0.359	30,940	1453	1.060
151	Coil_6	765	410	3.12	0.246	31,608	1315	0.877
152	Coil_6	1152	189	2.74	0.753	30,615	846	1.562
153	Coil_6	1152	290	3.45	0.522	36,120	998	1.897
154	Coil_6	1749	188	2.92	0.811	32,364	589	2.790
155	Coil_6	1749	311	3.38	0.416	33,168	604	1.241
156	Coil_6	1749	442	3.41	0.234	33,562	611	0.920

References

- [1] P. Naphon, S. Wongwises, A review of flow and heat transfer characteristics in curved tubes, *Renew. Sustain. Energy Rev.* 10 (5) (2006) 463–490.
- [2] S. Vashisth, V. Kumar, K.D. Nigam, A review on the potential applications of curved geometries in process industry, *Ind. Eng. Chem. Res.* 47 (10) (2008) 3291–3337.
- [3] A.M. Fsadni, J.P.M. Whitty, A review on the two-phase heat transfer characteristics in helically coiled tube heat exchangers, *Int. J. Heat Mass Transf.* 95 (2016) 551–565.
- [4] A.M. Fsadni, J.P.M. Whitty, A review on the two-phase pressure drop characteristics in helically coiled tubes, *Appl. Therm. Eng.* 103 (2016) 616–638.
- [5] G. Berthoud, S. Jayanti, Characterization of dryout in helical coils, *Int. J. Heat Mass Transf.* 33 (7) (1990) 1451–1463.
- [6] K.W. Hwang, D.E. Kim, K.H. Yang, J.M. Kim, M.H. Kim, H.S. Park, Experimental study of flow boiling heat transfer and dryout characteristics at low mass flux in helically-coiled tubes, *Nucl. Eng. Des.* 273 (2014) 529–541.
- [7] M.K. Jensen, A.E. Bergles, Critical heat flux in helically coiled tubes, *Trans ASME* 103 (1981) 660–666.
- [8] S. Jayanti, G. Berthoud, High - quality dryout in helical coils, *Nucl. Eng. Des.* 122 (1990) 105–118.
- [9] S. Jayanti, G.F. Hewitt, Prediction of onset of nucleate boiling, net vapour generation and subcooled CHF in coiled tubes, *Wärme-und Stoffübertragung* 26 (6) (1991) 301–305.
- [10] M. Kozeki, H. Nariiai, T. Furukawa, K. Kurosu, A study of helically coiled tube once-through steam generator, *Bull. Jpn. Soc. Mech. Eng.* 13 (60) (1970) 1485–1494.
- [11] H.C. Unal, M.L.G.G. Van, P.M.V. Verlaet, Dryout and two-phase flow pressure drop in sodium heated helically coiled steam generator tubes at elevated pressures, *Int. J. Heat Mass Transf.* 24 (1981) 285–298.
- [12] M.A. Styrikovich, V.S. Polonsky, V.V. Reshetov, Experimental investigation of the critical heat flux and post-dryout temperature regime of helical coils, *Int. J. Heat Mass Transf.* 27 (1984) 1245–1250.
- [13] W. Ma, M. Zhang, X. Chen, High-quality critical heat flux in horizontally coiled tubes, *J. Therm. Sci.* 4 (3) (1995) 205–211.
- [14] M. Kaji, K. Mori, T. Matsumoto, M. Oishi, T. Sawai, S. Nakanishi, Forced convective boiling heat transfer characteristics and critical heat flux in helically coiled tubes, 1998, pp. 3343–3349.
- [15] W.S. Chung, Y.C. Sa, J.S. Lee, An experimental study on dry out pattern of two-phase flow in helically coiled tubes, *KSME Int. J.* 16 (2002) 1540–1549.
- [16] G. Matsui, M. Suzuki, M. Kaji, Characteristics of boiling flow in helically coiled heat-transfer tube, *J. JSME* 8 (2008) 72–77.
- [17] T. Ami, N. Nakamura, T. Tsuruno, H. Umekawa, M. Ozawa, Boiling heat transfer and flow characteristics of liquid nitrogen in helically coiled tube, *Jpn. J. Multiphase Flow* 24 (2011) 567–576.
- [18] C.N. Chen, J.T. Han, T.C. Jen, L. Shao, W.W. Chen, Fluid-to-fluid modeling of two-phase flow critical heat flux in horizontal helically coiled tubes, *Nucl. Eng. Des.* 241 (2011) 1430–1437.
- [19] C.N. Chen, J.T. Han, T.C. Jen, L. Shao, W.W. Chen, Experimental study on critical heat flux characteristics of R134a flow boiling in horizontal helically-coiled tubes, *Int. J. Therm. Sci.* 50 (2011) 169–177.
- [20] C.N. Chen, J.T. Han, T.C. Jen, L. Shao, Dry-out CHF correlation for R134a flow boiling in a horizontal helically coiled tube, *Int. J. Heat Mass Transf.* 54 (2011) 739–745.
- [21] L.Z. Tan, J.T. Han, C.N. Chen, P.C. Dou, Studies on fluid-to-fluid modeling of critical heat flux in a helically coiled tube, *Adv. Mater. Res.* 588–589 (2012) 1777–1780.
- [22] Y.J. Chung, K.H. Bae, K.K. Kim, W.J. Lee, Boiling heat transfer and dryout in helically coiled tubes under different pressure conditions, *Ann. Nucl. Energy* 71 (2014) 298–303.
- [23] L. Santini, A. Cioncolini, C. Lombardi, M. Ricotti, Dryout occurrence in a helically coiled steam generator for nuclear power application, *EPJ Web of Conferences*, vol. 67, EDP Sciences, 2014, p. 02102.
- [24] B.K. Hardik, S.V. Prabhu, Boiling pressure drop and local heat transfer distribution of water in horizontal straight tubes at low pressure, *Int. J. Therm. Sci.* 110 (2016) 65–82.
- [25] D.C. Groeneveld, J.Q. Shan, A.Z. Vasić, L.K. Leung, A. Durmayaz, J. Yang, S.C. Cheng, A. Tanase, The 2006 CHF look-up table, *Nucl. Eng. Des.* 237 (15) (2007) 1909–1922.
- [26] B.K. Hardik, P.K. Baburajan, S.V. Prabhu, Local heat transfer coefficient in helical coils with single phase flow, *Int. J. Heat Mass Transf.* 89 (2015) 522–538.
- [27] M. Merilo, Fluid-to-fluid modeling and correlation of flow boiling crisis in horizontal tubes, *Int. J. Multiph. Flow* 5 (1979) 313–325.
- [28] P.K. Baburajan, G.S. Bisht, A.J. Gaikwad, S.K. Gupta, S.V. Prabhu, Measurement of steady-state CHF in horizontal channels for low-pressure, low-flow conditions, *Heat Transfer Eng.* 36 (4) (2015) 418–428.